

INTERFACE CONTROL DOCUMENT

System Communication Protocol

Version 2.0

1. SCOPE

This ICD defines the message formats and communication protocol between subsystems for the embedded platform (Version 2.0).

This version adds priority levels, CRC-16 checksums, and extended status reporting.

2. MESSAGE FORMAT

All messages use the following structure:

- msg_id : uint8 (message identifier)
- seq_num : uint16 (CHANGED: sequence counter 0-65535)
- priority : uint8 (NEW: 0=low, 1=normal, 2=high, 3=critical)
- payload_len: uint16 (length of payload in bytes)
- payload : uint8[] (up to 128 bytes, CHANGED from 64)
- crc16 : uint16 (CHANGED: CRC-16/CCITT replaces additive checksum)

3. MESSAGE IDENTIFIERS

MSG_ID_HEARTBEAT = 0x01 Periodic heartbeat
MSG_ID_STATUS = 0x02 System status report
MSG_ID_COMMAND = 0x03 Command message
MSG_ID_ACK = 0x04 (NEW) Acknowledgment message
MSG_ID_DIAGNOSTIC = 0x05 (NEW) Diagnostic data dump

4. STATUS REPORT PAYLOAD

The status report contains:

- system_id : uint8 (originating system)
- state : enum (IDLE=0, ACTIVE=1, ERROR=2, MAINTENANCE=3)
(CHANGED: added MAINTENANCE state)
- uptime_sec : uint32 (CHANGED: uint32 to support longer uptime)
- temperature : int16 (CHANGED: int16 for 0.1 degree resolution)
- voltage_mv : uint16 (NEW: supply voltage in millivolts)
- error_count : uint16 (NEW: cumulative error counter)

5. CHECKSUM

CRC-16/CCITT polynomial 0x1021, initial value 0xFFFF.

Computed over: msg_id, seq_num (2 bytes), priority, payload_len, and all payload bytes.

6. TIMING

Heartbeat interval: 500 ms (CHANGED from 1000 ms)
Status report interval: 2000 ms (CHANGED from 5000 ms)
Command timeout: 1000 ms (CHANGED from 2000 ms)
ACK timeout: 500 ms (NEW)

7. ACKNOWLEDGMENTS (NEW)

All commands (MSG_ID_COMMAND) must be acknowledged with MSG_ID_ACK within the ACK timeout. The ACK payload contains the original msg_id and seq_num of the command being acknowledged.